1. Data roles and validation

1. Fit, select, evaluate

Split the observations by **role before inspecting scores**:

- **training** fits each candidate;
- **validation** compares already-declared candidates;
- a **sealed final test set** is opened only after every model, hyperparameter, threshold, stopping rule, preprocessing choice, and reporting decision is fixed.

For disjoint training and validation index sets T, V, squared-loss validation for a candidate λ fixed before inspecting V is

$$\hat{R}_{V(\lambda)} = \frac{1}{|V|}\sum_{i\in V}\left(y_i - x_i^T\hat{\beta}_{\lambda,T}\right)^2.$$

Conditional on the fitted predictor and independent validation observations, this is a fresh-loss sample mean. If the same values choose $\hat{\lambda} = \operatorname{argmin}_\lambda \hat{R}_{V(\lambda)}$, then the selected minimum $\hat{R}_{V(\hat{\lambda})}$ is useful for selection but is **not** untouched final performance evidence. After selection, refit on the complete selection sample; evaluate the completed procedure outside the selection data.

2. k-fold cross-validation

Partition the available **selection sample** into disjoint folds $F_1, \ldots, F_k$. For candidate λ,

$$\mathrm{CV}_{k(\lambda)} = \frac{1}{n}\sum_{j=1}^{k}\sum_{i\in F_j}\ell\left(y_i, f_{-F_j}^{(\lambda)}(x_i)\right).$$

Each observation is scored exactly once by a fit that excludes its fold. Use the same fold partition across candidates, then choose λ and refit on the complete selection sample. Keep the final test set out of the fold table.

If fold sizes differ, average the **individual held-out losses** over all n observations. An unweighted mean of fold means gives small and large folds equal weight and generally differs from the displayed definition.

Changing k changes training size per fit, number of fits, overlap, and dependence among fold losses. Without extra assumptions, there is no universal monotone bias-variance ranking in k, and averaging k dependent fold results does not automatically divide a standard deviation by $\sqrt{k}$.

2. Leave one out

1. LOOCV definition and target

Leave-one-out cross-validation is k-fold CV with $k = n$. If $f_{-i}^{(\lambda)}$ is trained on every observation except i, then for squared loss

$$\mathrm{LOOCV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - f_{-i}^{(\lambda)}(x_i)\right)^2.$$

Every summand evaluates a procedure trained on $n-1$ observations, using the same candidate value and feature construction in every deletion fit. For a fixed dataset, singleton folds and the computed score are deterministic. Across new samples, the fitted models and score remain random. Transfer from the exact $n-1$-training target to the final n-observation fit requires stability or a sufficiently smooth learning curve near n.

2. Ridge response map

For fixed design $X \in \mathbb{R}^{n\times d}$ and fixed $\lambda > 0$,

$$\hat{\beta}_\lambda = \left(X^TX + \lambda I_d\right)^{-1}X^Ty, \quad \hat{y} = H_\lambda y,$$

$$H_\lambda = X\left(X^TX + \lambda I_d\right)^{-1}X^T \in \mathbb{R}^{n\times n}.$$

A rule $\hat{y} = Hy$ with H fixed with respect to y is a **linear smoother**. The diagonal $h_{\lambda,ii} = (H_\lambda)_{ii}$ is observation i's ridge leverage: the coefficient multiplying y_i in its own full-data fitted value. Keep the matrix H_λ distinct from a hypothesis class $\mathcal{H}$.

For full-rank OLS, $\mathrm{tr}(H_0) = \mathrm{rank}(X)$. For positive ridge,

$$\mathrm{tr}(H_\lambda) = \sum_j \frac{\mu_j}{\mu_j + \lambda},$$

over positive eigenvalues μ_j of X^TX. This is ridge **effective degrees of freedom** and is generally smaller than the unregularized rank.

3. Exact PRESS shortcut

1. Deleted residual from one full fit

Set

$$A_\lambda = X^TX + \lambda I_d, \quad e_i = y_i - \hat{y}_i.$$

Deleting row i changes the ridge normal equations by a rank-one term. Sherman-Morrison gives, when $1 - h_{\lambda,ii} \neq 0$,

$$\hat{\beta}_{-i,\lambda} = \hat{\beta}_\lambda - \frac{A_\lambda^{-1}x_ie_i}{1 - h_{\lambda,ii}},$$

$$\hat{y}_{-i,i} = \hat{y}_i - \frac{h_{\lambda,ii}e_i}{1 - h_{\lambda,ii}},$$

and therefore

$$e_{-i,i} := y_i - \hat{y}_{-i,i} = \frac{e_i}{1 - h_{\lambda,ii}}.$$

Thus, for compatible fixed-design ordinary or ridge least squares,

$$\mathrm{LOOCV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1 - h_{\lambda,ii}}\right)^2,$$

and PRESS is the corresponding unaveraged sum. One compatible full-data fit supplies all residuals and diagonal leverages. Correct each residual **before** squaring and averaging. High leverage can strongly inflate a small full-data residual after deletion.

2. Scope audit

The PRESS identity requires more than a full-data display $\hat{y} = Hy$:

- fixed design and the same feature construction after deletion;
- ordinary or quadratic-penalty least-squares normal equations;
- the same fixed penalty λ in every deletion fit;
- nonzero $1 - h_{\lambda,ii}$.

Full-data response-linearity alone does not reproduce case deletion. Ordinary k-nearest-neighbor prediction is a nonexample because deletion may change neighbor sets and weights. Do not assign it the least-squares PRESS formula.

Kernel ridge has compatible quadratic structure. With Gram matrix $K_{ij} = k(x_i, x_j)$,

$$M_\lambda = K(K+\lambda I_n)^{-1}, \quad \hat{y} = M_\lambda y.$$

For the linear kernel $K = XX^T$, the push-through identity gives $M_\lambda = H_\lambda$. Compatibility of the deletion structure—not the phrase "linear in y" alone—licenses the shortcut.

4. Information criteria

1. Likelihood convention and AIC

For independent pairs (X_i, Y_i), candidate M_j has regular conditional density $f_{j(y \texttt{ mid } x, \theta_j)}$, $\theta_j \in \mathbb{R}^{q_j}$, with

$$\ell_{j(\theta_j)} = \sum_i \log f_{j(y_i \texttt{ mid } x_i, \theta_j)}, \quad \hat{\theta}_j = \arg\max_{\theta_j} \ell_{j(\theta_j)}.$$

q_j counts **all** likelihood quantities fitted by MLE, including an estimated Gaussian variance. Ridge $\operatorname{tr}(H_\lambda)$ is not automatically this parameter count. Use one sign convention throughout:

$$D_{\text{fit},j} = -2\ell_{j(\hat{\theta}_j)}, \quad \text{AIC}_j = -2\ell_{j(\hat{\theta}_j)} + 2q_j,$$

with lower values preferred. The fresh-data target is expected conditional KL, equivalently expected negative conditional log score up to common constants. The $2q_j$ optimism correction requires a regular finite-dimensional conditional likelihood, MLE fitting, identifiable interior parameters, consistent counting, large-sample conditions, and the ordinary correctly specified information-equality case. AIC is not a distribution-free PAC bound; nonidentifiability or important misspecification needs separate theory.

A formal deviance $2\left(\ell_{\text{sat}} - \ell_{j(\hat{\theta}_j)}\right)$ ranks like -2ℓ only with a common saturated reference. The equivalent $2\ell - 2q$ form is **higher**-is-better: negate the entire criterion, never one term.

2. BIC and its identification boundary

The marginal likelihood integrates rather than maximizes:

$$p(y_{1:n} \texttt{ mid } x_{1:n}, M_j) = \int \exp\left(\ell_{j(\theta_j)}\right) \pi_{j(\theta_j)} \, d\theta_j.$$

For a regular identifiable model, interior MLE, positive-smooth prior near it, and positive-definite local curvature, Laplace concentration gives

$$-2\log p(y_{1:n} \texttt{ mid } x_{1:n}, M_j) = -2\ell_{j(\hat{\theta}_j)} + q_j \log n + O(1),$$

so $\text{BIC}_j = -2\ell_{j(\hat{\theta}_j)} + q_j \log n$, lower is better. BIC is a large-sample marginal-likelihood approximation under the declared model and prior regime. Unequal model priors also enter posterior odds; dropping $O(1)$ terms does not make the derivation prior-free.

Consistent identification additionally needs a declared candidate list containing the generating model, regular identifiable likelihoods, suitable positive prior mass, and a large-sample sequence. A finite-sample choice does not prove truth; the claim changes if truth is absent or the model is singular.

3. Target-assumption-guarantee map

Rule	Object	Supported reading
CV	held-out loss	fold-trained/selection-procedure performance, with qualified transfer
AIC	$-2\ell + 2q$	asymptotic fresh conditional KL/log score
BIC	$-2\ell + q\log n$	marginal-likelihood ranking; conditional identification
SRM	$\hat{R}_S + \text{class term}$	declared high-probability risk upper bound

LOOCV log score and AIC are not identical finite-sample formulas; AIC/BIC do not inherit SRM's PAC guarantee. Similar anti-overfitting roles do not create one interchangeable "training error + penalty" equation.

5. Honest evaluation

1. Select inside, evaluate outside

Final evidence must come from observations that influenced **no part** of the adaptive procedure. Use new/sealed data, or in each outer split:

1. fit and adapt only on the outer-training portion;
2. run the full inner rule (CV, AIC, BIC, or declared SRM);
3. return one completed procedure;
4. score it once on the untouched outer fold.

An inner-CV minimum is selection evidence; outer losses evaluate the complete selection algorithm. Never feed outer outcomes back into choice and then report their selected minimum. Information criteria may avoid a validation split, but cannot make training data fresh or license final-test reuse.

Choose by the complete brief: target, candidate family and regularity, available data, stability, refit cost, adaptation depth, and the outer boundary.

- ridge + squared loss + exact deletion structure: inner PRESS/LOOCV, then refit and outer/sealed evaluation;
- small regular MLE family + predictive log score: AIC or target-matched inner CV, plus untouched final log score if performance is reported;
- represented regular nested model + identification goal: BIC; any predictive claim still needs separate evaluation;
- adaptive trees/thresholds + classification loss: nested CV, with outer loss for the completed search.

"Use BIC when n is large" is incomplete: sample size alone does not specify the target, model family, assumptions, computation, adaptation, or data reuse.